Chapter 4: Random Variables (Sections 4.8 – 4.10)

4.8 Other Discrete Probability Distributions

4.8.1 The Geometric Random Variable

Represents the number of independent trials required to get the **first success**.

- **Definition:** Independent trials, each with success probability p. X is the trial number of the first success.
- **PMF:**

$$P(X = n) = (1 - p)^{n-1} p, \quad n = 1, 2, \dots$$

- **Properties:**
  - $\sum P(X = n) = 1$ (Success eventually occurs with probability 1).
  - **Expected Value:** $E[X] = \frac{1}{p}$
    - *Example:* Expected rolls of a fair die to get a 1 is $1/(1/6) = 6$.
  - **Variance:** $\text{Var}(X) = \frac{1-p}{p^2}$
  - **Tail Probability:** $P(X > k) = (1 - p)^k$ (Prob that first k trials are failures).

4.8.2 The Negative Binomial Random Variable

Represents the number of trials required to accumulate a total of r **successes**.

- **Definition:** Generalization of Geometric ($r = 1$). X is the trial number of the r-th success.
- **PMF:**

$$P(X = n) = \binom{n-1}{r-1} p^r (1 - p)^{n-r}, \quad n = r, r + 1, \dots$$

  - *Logic:* For the r-th success to happen on trial n, the first $n - 1$ trials must contain exactly $r - 1$ successes, and the n-th trial must be a success.
- **Relationship:** X can be written as the sum of r independent geometric random variables $X_1 + \cdots + X_r$.
- **Properties:**
  - **Expected Value:** $E[X] = \frac{r}{p}$
  - **Variance:** $\text{Var}(X) = \frac{r(1-p)}{p^2}$
- **Banach Match Problem:** A classic example involving two matchboxes, solved using negative binomial concepts to find the prob that the other box has k matches when one becomes empty.

4.8.3 The Hypergeometric Random Variable

Models sampling **without replacement** from a finite population.

- **Scenario:** Urn with N balls, m are white, $N - m$ are black. Select n balls. X is the number of white balls selected.
- **PMF:**

$$P(X = i) = \frac{\binom{m}{i}\binom{N-m}{n-i}}{\binom{N}{n}}, \quad i = 0, \dots, n$$

- **Properties:**
  - **Expected Value:** $E[X] = \frac{nm}{N} = np$ (where $p = m/N$).
    - *Note:* Same mean as Binomial, even though trials are dependent.
  - **Variance:** $\text{Var}(X) = np(1 - p)\left(\frac{N-n}{N-1}\right)$

- The term $\frac{N-n}{N-1}$ is the **finite population correction factor**. As $N \to \infty$, this approaches 1, and the variance approaches the Binomial variance.

4.8.4 The Zeta (Zipf) Distribution

Used to model size/frequency data (e.g., word frequency, city populations).

- **PMF:**

$$P(X = k) = \frac{C}{k^{\alpha+1}}, \quad k = 1, 2, \ldots$$

   - Where $\alpha > 0$ and $C = \left[\sum_{k=1}^{\infty} (1/k)^{\alpha+1}\right]^{-1}$ (related to the Riemann Zeta function).

4.9 Expected Value of Sums of Random Variables

One of the most useful properties in probability theory.

Linearity of Expectation

For any random variables $X_1, X_2, \ldots, X_n$ (dependent or independent):

$$E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i]$$

- *Proof:* Based on the fact that $E[X] = \sum_{s \in S} X(s)p(s)$. By reordering the summation over outcomes s, the sum of expectations emerges naturally.

Applications

1. **Sum of Dice:** Rolling n dice. $E[\text{Sum}] = n \times 3.5 = 3.5n$.
2. **Binomial Mean:** $X = \sum X_i$ (indicator variables). $E[X] = \sum p = np$.
3. **Hypergeometric Mean:** Even though trials are dependent (sampling without replacement), the probability of success on the i-th draw is still m/N. Thus $E[X] = n(m/N)$.
4. **Coupon Collector's Problem:** (From previous sections, usually solved using sums of Geometric variables).

Variance of Sums

Unlike expectation, variance is **not** generally linear.

$$E\left[\left(\sum X_i\right)^2\right] = \sum E[X_i^2] + \sum_{i \neq j} E[X_i X_j]$$

- If X_i are independent (or uncorrelated), cross terms simplify, and $\text{Var}(\sum X_i) = \sum \text{Var}(X_i)$.
- For Hypergeometric (dependent trials), the covariance terms lead to the finite population correction factor derived in section 4.8.3.

4.10 Properties of the Cumulative Distribution Function (CDF)

The CDF $F(b) = P(X \leq b)$ characterizes the random variable.

Properties

1. **Non-decreasing:** If $a < b$, then $F(a) \leq F(b)$.
2. **Limits:** $\lim_{b \to \infty} F(b) = 1$ and $\lim_{b \to -\infty} F(b) = 0$.
3. **Right Continuous:** $\lim_{n \to \infty} F(b_n) = F(b)$ for decreasing sequence $b_n \to b$.

Computing Probabilities from F

- $P(a < X \leq b) = F(b) - F(a)$
- $P(X < b) = \lim_{h \to 0^+} F(b-h)$ (Left limit)
- $P(X = b) = F(b) - \lim_{h \to 0^+} F(b-h)$ (Jump size at b)

Example

Given a step function $F(x)$:

- $P(X < 3)$ is the value of F just before 3.
- $P(X = 1)$ is the size of the jump at 1: $F(1) - F(1^-)$.
- $P(X > 1/2) = 1 - F(1/2)$.

Chapter Summary (Key Formulas)

Distribution	PMF $P(X=k)$	Mean $E[X]$	Variance $\mathrm{Var}(X)$
Binomial	$\binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$
Poisson	$e^{-\lambda} \lambda^k / k!$	λ	λ
Geometric	$(1-p)^{k-1} p$	$1/p$	$(1-p)/p^2$
Neg. Binomial	$\binom{k-1}{r-1} p^r (1-p)^{k-r}$	r/p	$r(1-p)/p^2$
Hypergeometric	$\frac{\binom{m}{k}\binom{N-m}{n-k}}{\binom{N}{n}}$	$\frac{nm}{N}$	$np(1-p)\frac{N-n}{N-1}$